

Influence of uveal melanoma therapy on patients' quality of life: a psychological study

Sophie Chabert, Michaela Velikay-Parel and Martin Zehetmayer

Department of Ophthalmology and Optometry, University of Vienna, Medical Faculty, Vienna, Austria

ABSTRACT.

Purpose: The objective of this study was to describe the quality of life of 98 patients treated with brachytherapy or stereotactic external beam irradiation, either with the Leksell Gamma Knife® or a 6-MV LINAC.

Methods: The EORTC QLQ-C30 and the EORTC QLQ-OPT37 psychological questionnaires and the HADS-D scale were used. Two visual analogue scales (VASs) were applied to compare overall quality of life before and after treatment.

Results: The results revealed no significant differences in quality of life ($p = 0.215$) among treatments with different methods of radiotherapy. Comparison of quality of life before and after therapy did, however, indicate a decrease in quality following radiotherapy. The average binocular visual acuity (VA) was 0.8 (range 0.3–1). Complications arising from the treatment included a reduction in VA in the affected eye, keratitis, cataract, scleral and corneal necrosis, radiation retinopathy, radiation optic neuropathy, retinal detachment and glaucoma.

Conclusion: Due to the diagnosis revealed and the subsequent radiotherapy, patients with uveal melanoma experienced a reduced quality of life. However, compared to patients with other types of cancer, they seemed to feel relatively well and showed fewer signs of deterioration.

Key words: quality of life – uveal melanoma – anxiety – depression – psychological study

Acta Ophthalmol. Scand. 2004; 82: 25–31

Copyright © Acta Ophthalmol Scand 2003.

doi: 10.1046/j.1600-0420.2003.00210.x

Introduction

Over the past two decades, research into quality of life has become an integral part of clinical trials and the importance of measuring it has been increasingly recognized. The aim of this research is to identify and better describe the damaging effects of the disease in question and its treatment; thus an assessment of quality of life is considered to be an important evaluation criterion in making medical deci-

sions and ultimately constitutes a vital factor in achieving success in therapy.

Research into the domain of quality of life makes it possible to determine the significance of the various advantages and disadvantages of a particular cancer therapy and to establish which of the therapies now available will be likely to have the least harmful effects on the quality of life of a patient. Research thus assists both doctor and patient in choosing among different treatment options.

Concerning patients with uveal melanoma, quality of life has remained largely unexplored and therefore relatively little is known about it. Only a few reports deal with the experience of patients with uveal melanoma (Manschot et al. 1996; Damato 2000; Foss et al. 2000).

In order to examine this interesting subject, we launched a psychological study to measure quality of life in a group of patients conservatively treated for uveal melanoma with different methods of radiotherapy.

Material and Methods

Patient population

The study included 130 patients with uveal melanoma, treated with different methods of radiotherapy such as stereotactic external beam irradiation (SEBI) with a 6-MV LINAC, the Leksell Gamma Knife® or Ruthenium-106 brachytherapy. The methods of treatment have been described previously (Lommatzsch 1987; Zehetmayer et al. 2000; Dieckmann et al. 2001).

The medical records of all patients diagnosed with uveal melanoma at the Department of Ophthalmology, Vienna University Hospital, during the years 1988–2001 were reviewed. To be eligible for the study, patients had to have been treated with one of the methods mentioned above.

Ruthenium-106 brachytherapy was offered to patients with uveal melanoma

of <7 mm thickness and >3 mm distance to the optic disc and macula.

Stereotactic external beam irradiation was offered only to patients with uveal melanoma which were deemed unsuitable for Ruthenium-106 brachytherapy or local resection because of melanoma of >7 mm tumour height or juxtapapillary and/or juxtamacular tumours (height >3 mm). It was also offered to patients with a posterior tumour margin within 3 mm distance of the optic disc and/or fovea.

Patients were excluded if they presented evidence of echographic extra-scleral tumour extension or neovascular glaucoma or had submitted to any kind of pretreatment or metastasising at baseline.

Patients whose residence was outside Austria were also excluded from the study sample.

Data survey and collection

This study was conducted retrospectively at the Department of Ophthalmology, University Hospital Vienna.

Clinical data were gathered from the original medical records. Data regarding quality of life were collected with the help of questionnaires.

During June–September 2001 questionnaires were mailed to all patients, along with a return envelope and an explanatory letter.

The cover letter detailed the purpose of the study and explained that participation was entirely voluntary. Patients were also informed that the results of the study would be compiled by the research team and that individual responses would remain anonymous. Patients were therefore encouraged to answer as truthfully as possible.

Each questionnaire was labelled with a specific number so that results could be correlated with clinical data. Time between treatment and completion of the questionnaires was about 3 years.

A total of 98 patients returned the questionnaires duly completed. Ten of 130 patients had died and another 11 had moved elsewhere and could not be traced. Only 10 patients refused to participate, yielding a total compliance rate of 90%.

Questionnaires

To compare quality of life before and after treatment, we used two visual analogue scales (VASs). Other data were

collected using the EORTC QLQ-C30 questionnaire (Version 3) (Aaronson et al. 1993), the EORTC QLQ-OPT37 questionnaire (Brandberg et al. 2004) and the HADS-D scale (Herrmann et al. 1995), as well as another form for collecting socio-demographic and medical data.

Both EORTC questionnaires are used to collect data on the quality of life of cancer patients. The QLQ-C30 (main questionnaire) concentrates on problems related to various specific forms of cancer, whereas the QLQ-OPT37 is specially designed for patients suffering from eye tumours (annexed questionnaire).

EORTC QLQ-C30

The EORTC QLQ-C30 questionnaire consists of 30 items. Items 1–28 relate to physical symptoms of illness, while items 29 and 30 require overall information on health and quality of life.

All items can be allocated to different scales. The questionnaire incorporates five functional scales (physical, role, cognitive, emotional, social), three symptom scales (fatigue, pain, nausea and vomiting), a number of individual item scales and a global health status, as well as a quality of life scale.

Questions on symptoms (items 1–28) are designed in categorical form and require answers to be marked on a scale of 1–4, where 1 = not at all and 4 = very much. Questions concerning global health and quality of life require answers to be marked on a scale of 1–7, where 1 = very bad and 7 = excellent. The questionnaire is easy to score, encompassing both an overall score and scores for each subscale. All scores can be transformed to continuous scales ranging from 0 to 100, with the higher scores representing better quality of life; exceptions are the symptom scales, where higher scores indicate more severe symptoms.

The questionnaire is validated and widely used; it has been described in detail elsewhere (Aaronson et al. 1993).

EORTC QLQ-OPT37

The EORTC QLQ-OPT37 questionnaire is a provisional module designed by the Ocular Oncology Task Force in collaboration with the Quality of Life Group of the European Organization for Research and Treatment of Cancer (EORTC). The questionnaire deals with eye symptoms (e.g. ocular gritti-

ness, ache, watering, dryness, itching), vision (floaters, interference with the other eye), problems with appearance, worries about recurrent disease, specific functional problems due to vision impairment (e.g. reading, driving, pouring tea, watching television, taking steps, headache). It consists of 37 items (with item numbers 31–67). Questions are designed in a categorical form and require answers to be marked on a scale of 1–4, where 1 = not at all and 4 = very much. The questionnaire yields an overall score and scores for each subscale. Scores cannot as yet be transformed to continuous scales because unfortunately no standards exist up to now; however, work is underway to establish them and thus check reliability and validity.

The appropriateness of the items was checked with a number of patients who were interviewed in phase 2 of the development process. The final module (OPT30) is now complete for phase 4 testing (Brandberg et al. 2004).

At the time of our study, no other specific questionnaire was available for this group of patients. Of course, a number of generic instruments such as the EORTC QLQ-C30 and the HADS have been developed to measure how disease and treatment influence the ways in which patients continue to survive physically, psychologically and socially. However, these generic questionnaires investigate general factors only (fatigue, anxiety, depression, family life, social activities, financial difficulties) and cannot show disease-specific symptoms. Therefore, a dedicated instrument to address the specific symptoms and difficulties caused by this particular disease was required. The EORTC QLQ-OPT37 questionnaire is such a specific instrument. Thus, the initial QLQ-OPT37 was our choice.

HADS (Hospital Anxiety and Depression Scale)

The HADS-D (hospital anxiety and depression scale; D = German version) is a short, self-assessment form used to analyse the psychological variables anxiety and depression in somatic medicine.

It consists of 14 items, seven of which refer to anxiety and seven to depression.

The anxiety scale of the HADS form is designed to describe psychological manifestations of anxiety neuroses exclusively. The information compiled deals predominantly with a generalized

anxiety symptomatology. The items of the depression scale are based on the concept of a mild, so-called 'endogenous' symptomatology.

The HADS scale is easy and quick to administer and has now become a well-established research tool. The scale is often used in somatic medicine and was chosen for its use in identifying and monitoring the psychological status of patients under treatment for cancer. As the HADS seems to be a useful tool for measuring the psychological dimension of the quality of life, it was used in this study.

The answers to every question are scored on a scale of 0–3, where 0 = not at all and 3 = a lot. The sum of the individual scores yields an overall depression and anxiety score. The highest possible number of points is 21. Scores higher than 10 on either subscale are considered to indicate morbidity, scores between 8 and 10 are interpreted as borderline levels of anxiety or depression, and scores below 8 are not significant and considered normal (Herrmann et al. 1995); thus a cut-off point of 10/11 seems to indicate the borderline between patients who are coping well with their cancer and those who have developed morbid anxiety and depression.

Visual analogue scales (VASs) to compare quality of life before and after treatment

Visual analogue scales require patients to rate a 10-cm line to indicate how they would judge their quality of life as it was before therapy and on this day after radiotherapy. Each line ranges from 0 to 100%.

Quality of life VASs were chosen because questions based on a VAS have been demonstrated to provide a greater range of patient response and have been used successfully for measuring subjective impressions. For these reasons, we included VAS scales, despite the availability of the QLQ-C30.

Statistical analysis

Statistical analysis was effected using spss for Windows, Version 10.0.

Demographic and clinical characteristics, including visual acuity (VA) outcome and complications arising from the treatment, were summarized and set down in a descriptive manner.

Scores from the questionnaires EORTC QLQ-C30 and EORTC

QLQ-OPT37 were shown as the mean ($\pm$ SEM). This was also applied to scores from the HADS.

Means were also calculated for VAS scores.

Analysis of association or differences between variables were performed by using non-parametric methods as the distribution of most variables was skewed. A p-value of 0.05 or less was considered significant.

The Mann–Whitney Test was employed for analysing the association between quality of life and various methods of treatment.

Differences among methods of radiotherapy related to physical and psychological symptoms were tested by the Kruskal–Wallis rank sum test.

Results

Demographic details

The study included 98 patients (51 male and 47 female), aged between 36 and 89 years, with an average age of 63 years. A total of 82.6% ($n=81$) of the patients were married or cohabiting, 4% ($n=4$) were single, 1% ($n=1$) separated and 12.4% ($n=12$) widowed.

Most of the patients (60.2%, $n=60$) were retired. However, 28.6% ($n=28$) were working at home or had regular out-of-house employment, 3% ($n=3$) were unemployed, 5.1% ($n=5$) were on certified sick leave and only 2.1% ($n=2$) were unable to work due to their state of health.

Clinical details

Exactly 50% of the patients ($n=49$) were treated with LINAC, while 19.4% ($n=19$) were treated with Ruthenium-106 and 21.4% ($n=21$) with Gamma Knife®. Seven patients (7.1%) underwent secondary enucleation because of treatment failure or complications.

When asked to describe their level of satisfaction with the treatment and care they received in hospital, 84.7% ($n=83$) of the patients said that they had been well informed about their illness and 84.7% ($n=83$) stated that they had also been adequately informed about the various treatments and their respective consequences. Almost all the patients (93.7%, $n=89$) stated that they had been involved in the choice of therapy. A total of 76.3% ($n=74$) said that they had been given the opportunity

to speak with doctors and/or members of the nursing staff about their problems during hospitalization.

In summary, patient responses indicated that the medical information and support provided by doctors and nursing staff during hospitalization and follow-up treatment were considered satisfactory.

At the end of the study, VA outcome and complications arising from the treatment were examined.

For the majority of patients, treatment of the eye by irradiation resulted in a marked diminishing of VA in the affected eye. In 25% of the patients, intensive irradiation led to a complete loss of visual function.

Other complications arising from the treatment were keratitis, cataract, scleral and corneal necrosis, radiation retinopathy, radiation optic neuropathy, retinal detachment and glaucoma.

The average binocular VA was 0.8 (range 0.3–1).

Quality of life

Of the 98 patients who completed the questionnaires, 18 (18.4%) declared that their quality of life had not changed as a result of radiotherapy, 48 (49%) stated that their quality of life had been reduced, 25 (25.5%) felt they had achieved better quality of life after radiotherapy, and seven (7.1%) did not reply to this question.

Comparison of levels of quality of life before and after treatment with radiotherapy generally revealed that quality of life deteriorated after radiotherapy. The level of quality of life as individually experienced by patients reached a mean of 74.1% before therapy and a mean of 69% after therapy, indicating a decline in quality of life of only 5% (Fig. 1).

Physical symptoms and side-effects of therapy, measured with the EORTC QLQ-C30 and the EORTC QLQ-OPT37 questionnaires

Many symptoms indicated a declining quality of life.

Quality of life scores derived from the QLQ-C30 questionnaire

The QLQ-C30 items 1–28 require answers on a scale of 1–4, with higher scores representing more pronounced symptoms. Results showed means of between 1 and 2; only item 22

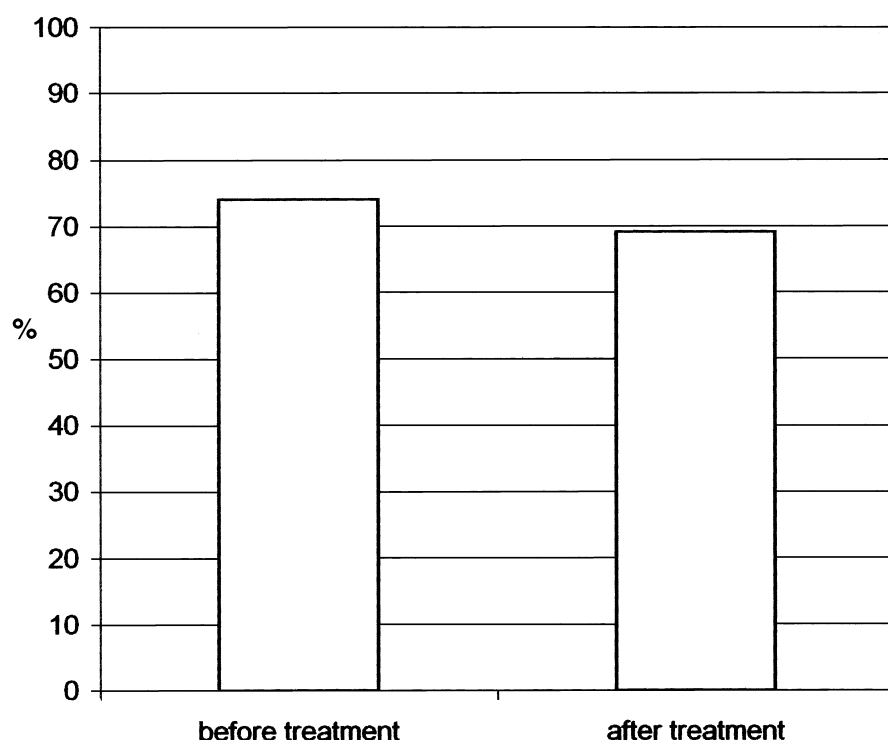


Fig. 1. Quality of life (%) before and after treatment with radiotherapy. Quality of life can range from 0 to 100. Higher scores represent a higher quality of life.

(‘Did you get worried?’) and item 7 (‘Did you feel limited in terms of hobbies and spare time activities?’) had means of just over 2. The response scale for items 29 (‘How would you rate your health status during the past few weeks?’) and 30 (‘How would you rate the level of your quality of life in

the past few weeks?’) range from 1 = very bad to 7 = excellent. Higher scores represent a better quality of life. We recorded a mean value of 5 for both items, implying that the quality of life and state of health experienced by patients both seemed to be rather good.

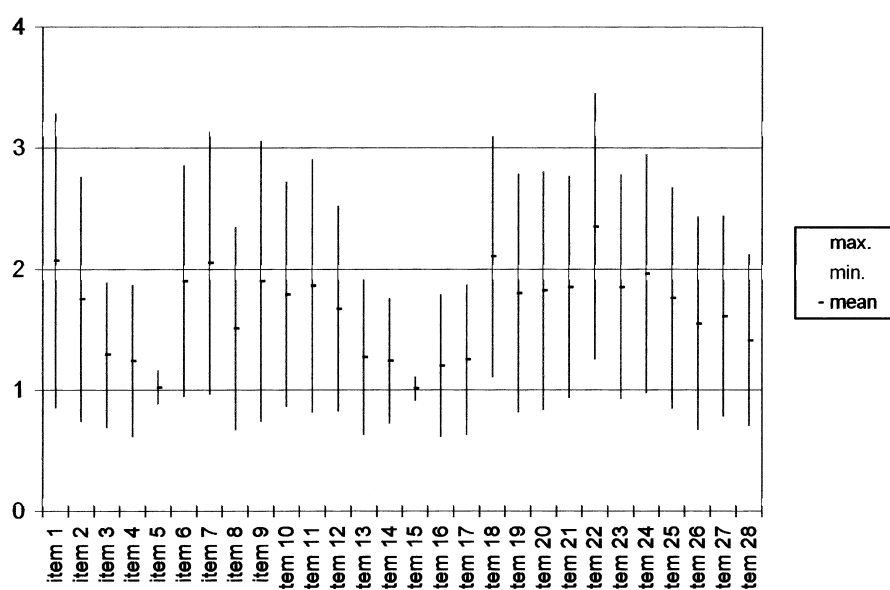


Fig. 2. EORTC QLQ-C30 questionnaire: results of single items. Data are shown as mean values ($\pm$ SEM). Scores deriving from items 1–28 range from 1 to 4, with higher scores representing more pronounced symptoms.

Means derived from this questionnaire are shown in Fig. 2 (single items) and Fig. 3 (subscales).

Quality of life scores derived from the QLQ-OPT37 questionnaire

The results generated by the QLQ-OPT37 yielded means slightly higher than those yielded by the QLQ-C30. On average, we found mean values between 2 (low) and 3 (moderate) or just under 2. Moderate problems were experienced during reading (2.86). The results showed that there was little to moderate deterioration due to the side-effects of therapy. The quality of life scores derived from the QLQ-OPT37 are shown in Fig. 4 (single item results) and Fig. 5 (subscale results).

Data are presented as the mean for each item and for each subscale. Scores range from 1 to 4, with higher scores representing a major manifestation of symptoms.

Anxiety and depression as a result of the psychological effects of illness and of the side-effects of therapy

Individual HADS items (possible scores 0, 1, 2, 3) varied between a minimum of 0.45 (items 12, 14) and 1.15 (item 11).

The mean scores of the HADS were 5.67 for anxiety ($SEM \pm 4.01$) and 4.93 for depression ($SEM \pm 4.25$). Scores of 8 or more on a subscale are taken to indicate possible pathology. Using this as a cut-off point, 27.9% of the patients ($n=27$) were within the range for borderline and clinical anxiety and 23.7% ($n=23$) for borderline depression. The percentages of patients with normal borderline or morbid anxiety and depression scores are shown in Table 1.

Severe anxiety scores were recorded for 12.4% of patients. We did not find any therapy-specific differences.

Scores indicating severe depression were recorded for 14.4% of patients. They also had abnormal anxiety scores.

Differences among various kinds of therapy

The question of interest in this context concerned whether differences exist between the types of therapy, with particular reference to side-effects and symptoms after treatment. In order to resolve this question, symptoms experienced by patients were compared and measured by using questionnaires for

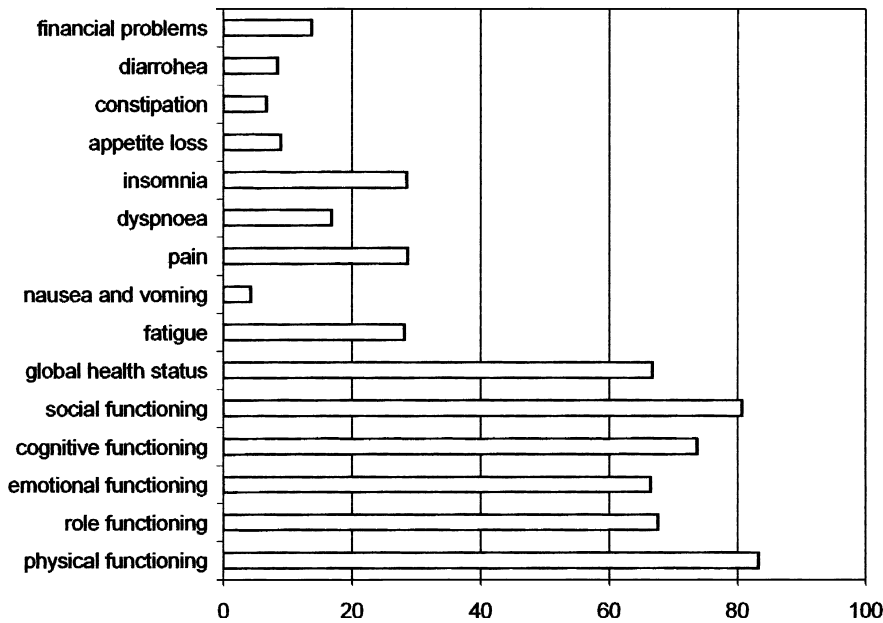


Fig. 3. EORTC QLQ-C30 questionnaire: results of subscales. Data are shown as mean values. Scores have been transformed to 0–100 scores, with higher scores representing better quality of life except for the symptom scales, where higher scores represent a major manifestation of symptoms.

each therapy subgroup. Results were as follows.

Scores taken from the HADS questionnaire did not show major differences among the types of therapy used, implying that the method of radiotherapy did not have a direct bearing on anxiety and depression experienced by patients. With one exception, this also applied to all QLQ-C30 items and almost all OPT37 items. Only a single QLQ-C30 item and six QLQ-

OPT37 items differed significantly (Table 2).

Based on the results of all QLQ-C30 and QLQ-OPT37 items, no major difference corresponding to type of radiotherapy concerning life quality after therapy was detectable ($p = 0.215$). Therefore, quality of life after treatment did not appear to depend on radiotherapy;

Moreover, we cannot conclude that any of the three types of therapy

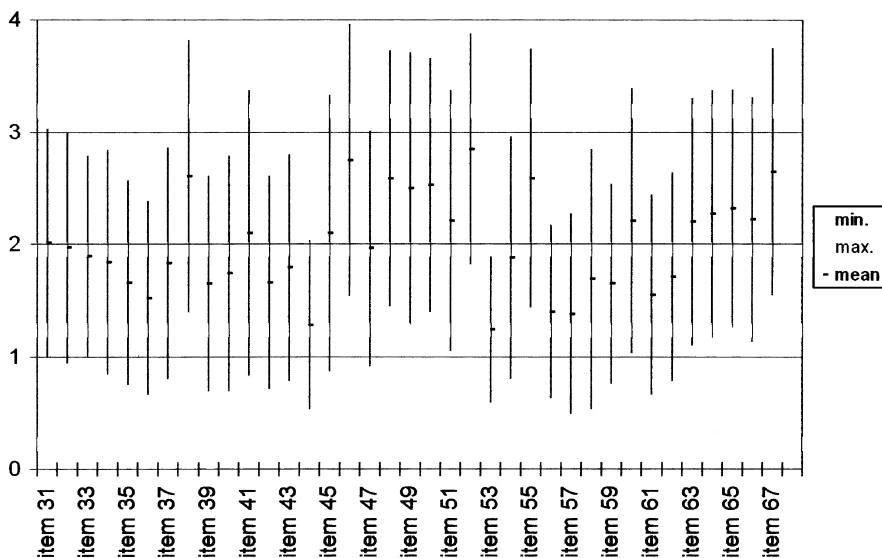


Fig. 4. EORTC QLQ-OPT37 questionnaire: results of single items. Data are shown as mean values ($\pm$ SEM). Scores range from 1 to 4, with higher scores representing more severe symptoms.

(LINAC, Gamma Knife[®] or Ruthenium-106) is significantly better or has fewer side-effects than the others.

Finally, when we compared enucleated cases with those treated conservatively, results revealed no significant differences in quality of life and post-therapeutic symptoms. Only responses to item 49 ('Were you troubled about tumour recurrence in the treated eye?') differed between these two groups of patients; the mean was markedly higher in the enucleated group than in the group of patients treated with LINAC, Gamma Knife[®] or Ruthenium-106.

Discussion

There are few mentions in the literature about conservative management of the post-therapeutic physical and psychological stress caused by the treatment as such.

Thus, the purpose of the current study was to document the quality of life reported by patients suffering from uveal melanoma.

Our results revealed that patients had a very nearly consistent level of quality of life, with a mean of approximately 70%, as measured with two visual analogue scales. After treatment with radiotherapy, quality of life deteriorated by about 5%. Loss of vision, loss of the eye, pain, uncertainty about the course of the disease, fear of a recurrent tumour and of death by metastatic diseases may be reasons for the reduced quality of life after treatment, not to mention all the secondary consequences of these problems, such as disruption of normal daily activities and responsibilities.

The risk of damage from radiotherapy is generally associated with the physical properties of the radiation and depends on the type of radiation used, its energy, the spatial dose distribution and the irradiated volume.

Patients with uveal melanoma may be treated by a number of modalities. The success of the treatment depends not only on the medical therapy but also on how well the medical team is able to restore the patient's peace of mind. It is very important that we establish how disease and treatment influence the way in which patients continue to survive physically, psychologically and socially. A key factor in

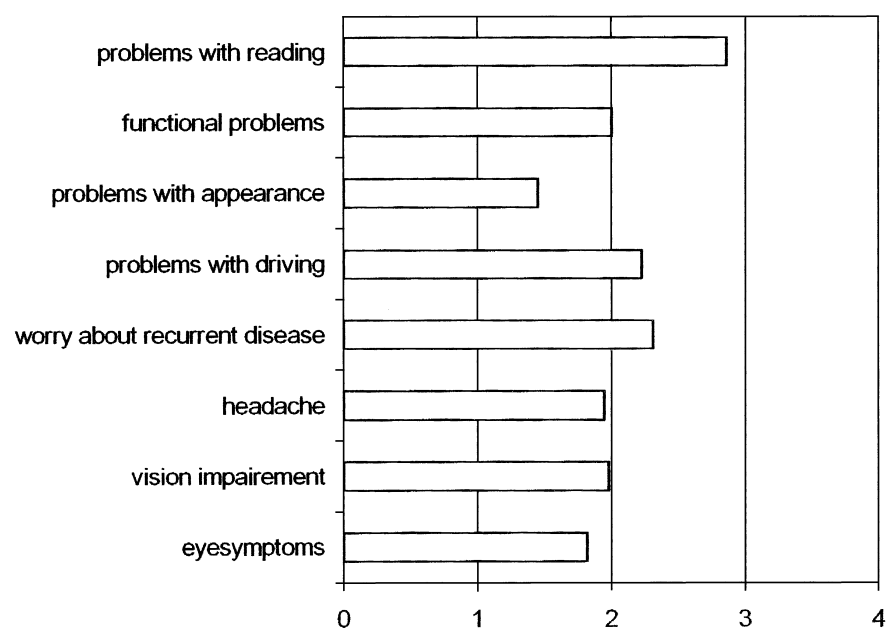


Fig. 5. EORTC QLQ-OPT37 questionnaire: results of subscales. Data are shown as mean values. Scores range from 1 to 4. Higher scores represent a major manifestation of symptoms.

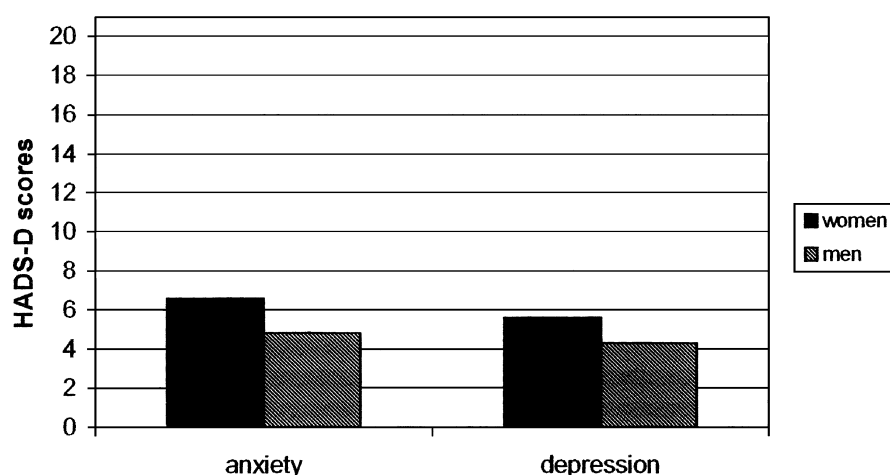


Fig. 6. Hospital Anxiety and Depression Scale (HADS-D). Levels of anxiety and depression in men and women after radiotherapy.

selecting the appropriate treatment concerns its likely impact on the patient's quality of life.

When comparing the different kinds of treatment with radiotherapy, neither of the two EORTC questionnaires showed major differences with respect to symptoms and side-effects of therapy. Only in seven of 56 items did we

notice significant differences between people treated with LINAC and those treated with Ruthenium-106 or Gamma Knife®.

The LINAC group experienced more soreness or discomfort in or around the treated eye and cold wind caused more pain. Differences in patient population but also in radiation distribution may

be a possible explanation. Patients treated with LINAC also had more difficulties in reading and watching television and objects appeared more distorted. This group comprised relatively more patients who had been treated for centrally located tumours.

When comparing patients treated with radiotherapy with those who had undergone secondary enucleation, results showed that the latter reported more fears of tumour recurrence. A similar result was found in a preliminary study in a sample of 240 patients with treated uveal melanoma, where patients were as likely to report fears of local tumour recurrence after enucleation as after conservative therapy (Damato 2000). This contrasts with results published by Manschot et al. (1996), where patients treated with primary enucleation were found to be unaware of the potential anxiety and depression that might be caused by an intraocular malignancy that might disseminate and cause a much-feared metastatic death. This could be due to differences in patients' points of view after primary and secondary enucleation.

In our sample, the impact of enucleation on the post-therapeutic quality of life was not found to be significant. Similar results in four vision-dependent activities (visual working, driving, reading and television viewing) were reported by Edwards & Schacht (1991).

In general, patients reported few physical symptoms. Looking at the subscale results, we observed that patients experienced problems with reading. Worries about recurrent diseases were also pronounced more often. Only a few people demonstrated worries about their appearance. Physical and social functioning were well developed. Symptoms such as constipation, diarrhoea, appetite loss, nausea and vomiting hardly ever occurred. Patients were disturbed by insomnia and pains. The present results obtained with the HADS-D scale indicated that severe and borderline anxiety manifested in 27.9% of patients with uveal melanoma. Women showed higher levels of anxiety than men, but there was no gender-specific difference with respect to depression. We observed morbid and borderline depression in 23.4% of patients. In summary, patients with uveal melanoma experienced anxiety rather than depression

Table 1. Patients (%) with normal, borderline or morbid anxiety and depression.

	Anxiety		Depression	
7 = not significant	72.12%	(n = 70)	76.3%	(n = 74)
8–10 = borderline	15.5%	(n = 15)	9.3%	(n = 9)
11 = pathological	12.4%	(n = 12)	14.4%	(n = 14)

Table 2. The mean rank for each item and for each therapy; higher mean ranks represent more pronounced symptoms.

Item no.	Therapy		
	Ruthenium-106	Gamma Knife®	LINAC
QLQ-C30-3	57.67	48.09	43.79
QLQ-OPT37 Item 32	37.06	38.32	53.23
Did you feel any pain, soreness or discomfort in or around the treated eye? (p = 0.013)			
QLQ-OPT37 Item 34	2.56	40.77	52.58
Did cold wind cause any pain in the treated eye? (p = 0.07)			
QLQ-OPT37 Item 41	42.47	31.38	45.57
Did things appeared distorted out of your treated eye? (p = 0.053)			
QLQ-OPT37 Item 43	51.06	27.95	46.42
Did you see floaters with your treated eye? (p = 0.002)			
QLQ-OPT37 Item 59	43.64	35.66	51.62
Did you have difficulty seeing well enough to watch television? (p = 0.034)			
QLQ-OPT37 Item 67	31.87	48.45	51.85
Did you have difficulty with reading because of your vision? (p = 0.016)			

(see Fig. 6). This result is unsurprising as considerable anxiety may arise from frequent control examinations, treatment after recurrence and treatment of complications after radiotherapy.

The HADS scale was used by another group of investigators (Maraste et al. 1992) to assess feelings of anxiety and depression in women who had been surgically treated for early breast cancer and referred for adjuvant radiotherapy. Their results were similar to ours and showed that the emotional distress of the women was characterized by anxiety rather than by depression. Validation of this HADS anxiety score against psychiatric ratings has indicated that there is a great likelihood that the scale detects anxiety, whereas the predictive capacity of the depression scale is not as pronounced (Hopwood et al. 1991). This may be another explanation for our results. Accordingly, the high proportion of patients experiencing morbid and borderline anxiety seems to be realistic; it is possible that the incidence of depression was underestimated in our study.

Last but not least, the results of the EORTC questionnaires and the HADS scale appear to indicate that radiotherapy slightly affects the quality of life of patients with uveal melanoma, but they appear to feel relatively well and show fewer signs of deterioration compared to patients suffering from other types of cancer. The rather high level of quality of life as expressed by the patients (almost 70% after treatment) may also be explained by the fact that their binocular VA (at 0.8) was still relatively good.

References

- Aaronson NK, Ahmedazai S, Bergman B, Bullinger M, Cull A & Duez NJ (1993): A quality of life instrument for use in international clinical trials in oncology. *J Natl Cancer Inst* **85**: 365–376.
- Brandberg Y, Damato BE, Kivelä T, Kock E & Seregard S (2004): The development and pre-testing of the EORTC ophthalmic quality of life questionnaire module (OPT-30). *Eye*, in press.
- Damato B (2000): Disease and 'dis-ease' in patients with uveal melanoma. *Br J Ophthalmol* **84**: 343–345.

- Dieckmann K, Bogner J, Georg D, Zehetmayer M, Kren G & Potter R (2001): A linac-based stereotactic irradiation technique of uveal melanoma. *Radiother Oncol* **61** (1): 49–56.
- Edwards MG & Schacht AR (1991): Impact of enucleation for choroidal melanoma on the performance of vision-dependent activities. *Arch Ophthalmol* **109**: 519–521.
- Foss AJE, Lamping DL, Schroter S & Hungerford J (2000): Development and validation of a patient-based measure of outcome in ocular melanoma. *Br J Ophthalmol* **84**: 347–351.
- Herrmann CH, Buss U & Snaith RP (1995): HADS-D Hospital Anxiety and Depression Scale – Deutsche Version. [Ein Fragebogen zur Erfassung von Angst und Depressivität in der somatischen Medizin.] Bern, Göttingen, Toronto, Seattle: Hans Huber.
- Hopwood P, Howell A & Maguire P (1991): Screening for psychiatric morbidity in patients with advanced breast cancer: validation of two self-report questionnaires. *Br J Cancer* **64**: 353–356.
- Lommatzsch PK (1987): Results after beta-irradiation (106Ru/106Rh) of choroidal melanomas. Twenty years' experience. *Am J Clin Oncol* **10** (2): 146–151.
- Manschot WA, Lee WR & van Strik R (1996): Uveal melanoma: updated considerations on current management modalities. *Int Ophthalmol* **19**: 203–209.
- Maraste R, Brandt L, Olsson H & Ryde-Brant B (1992): Anxiety and depression in breast cancer patients at start of adjuvant radiotherapy. *Acta Oncol* **31**: 641–643.
- Zehetmayer M, Kitz M, Menapace R et al. (2000): Local tumour control and morbidity after one to three fractions of stereotactic external beam irradiation for uveal melanoma. *Radiother Oncol* **55** (2): 135–144.

Received on January 10th, 2003.
Accepted on September 13th, 2003.

Correspondence:

Martin Zehetmayer MD
Department of Ophthalmology and Optometry
Medical School
University of Vienna
General Hospital of Vienna
Währingergürtel 18–20
A-1090 Vienna
Austria
Tel: +43 1 40 400 7900
Fax: +43 1 40 400 7902
Email: martin.zehetmayer@akh-wien.ac.at